

Junhan Kong Curriculum Vitæ

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BIO

Junhan “Judy” Kong is a Ph.D. candidate in the Information School at the University of Washington. She is advised by Prof. Jacob O. Wobbrock and is a member of the DUB Group. She obtained her bachelor’s and master’s degrees in computer science from Carnegie Mellon University with an additional major in human-computer interaction (HCI) and minors in machine learning and statistics. Her research interests include **HCI, accessibility, human-AI interaction, adaptive interfaces, and AR/VR**. Her work seeks to enable computer technologies to understand the varying abilities of their users, and to design, implement, and evaluate tools to make technologies accessible by adapting to these abilities. Specifically, her work focuses on:

- Computational understanding of human abilities through user input behavior and sensor data;
- Mixed-initiative interface adaptations based on varying user abilities and contexts.

EDUCATION

University of Washington, Seattle WA Sep 2020 - Jun 2027 (expected)

Ph.D. in Information Science

Thesis (proposed): Intelligent and Customizable Interfaces for Ability-Aware Accessibility

Advisor: Jacob O. Wobbrock

Carnegie Mellon University, Pittsburgh PA May 2019 - May 2020

M.S. in Computer Science

Thesis: An Authoring Tool for Creating Interactive AR User Tutorials by Demonstration

Advisor: Jeffrey P. Bigham

Carnegie Mellon University, Pittsburgh PA Aug 2015 - May 2019

B.S. in Computer Science

Additional major in Human-Computer Interaction, minors in Statistics and Machine Learning

AWARDS AND HONORS

Best Paper Award, MobileHCI 2024

Best Paper Nominee, ASSETS 2022

Special Recognitions for Outstanding Reviews, CHI 2022, UIST 2023 & 2024

Adobe Research Intern Project Expo Winner, Adobe Intern Project Expo 2023

Boeing Blue Skies Award: Game Changer, CMU Undergraduate Research Symposium 2019

University Honors for Academic Excellence, Carnegie Mellon University

Dean’s List, Carnegie Mellon University, School of Computer Science

Best Educational App, TartanHacks (CMU Hackathon) 2017

Social Impact Prize, TartanHacks (CMU Hackathon) 2016

PUBLICATIONS

Peer-Reviewed Conference Papers

[C.9] **Junhan Kong**, Jacqui Fashimpaur, Michael J Proulx, Hemant Bhaskar Surale, Jacob O. Wobbrock, Amy Karlson. Quantifying 3D Pointing: Characterizing What Happens During Pointing Selection in Virtual Reality. In Proceedings of the ACM International Conference on Multimodal Interaction (ICMI '26), October 05–09, 2026, Napoli, Italy. <https://doi.org/10.1145/3776574.3831138>. (To Appear)

[C.8] Claire L. Mitchell, **Junhan Kong**, Jesse J. Martinez, Shaun K. Kane, Amy J. Ko, Alexis Hiniker, Jacob O. Wobbrock. Ability Heuristics for Conducting Accessibility Inspections. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3772318.3791504>.

[C.7] Xiyuan Shen*, Seokhyun Hwang*, **Junhan Kong**, Alexandre L. S. Filipowicz, Andrew Best, Jean M. Costa, Scott Carter, James Fogarty, Jacob O. Wobbrock. Touchscreens in Motion: Quantifying the Impact of Cognitive Load on Distracted Drivers. In the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25), September 28–October 1, 2025, Busan, Korea. <https://doi.org/10.1145/3746059.3747683>.

[C.6] **Junhan Kong**, Jacob O. Wobbrock, Tianyuan Cai, Zoya Bylinskii. Supporting Mobile Reading While Walking with Automatic and Customized Font Size Adaptations. In Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI '25), April 26–May 1, 2025, Yokohama, Japan. <https://doi.org/10.1145/3706598.3713367>.



[C.5] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. The Ability-Based Design Mobile Toolkit (ABD-MT): Developer Support for Runtime Interface Adaptation Based on Users' Abilities. In Proceedings of the ACM on Human-Computer Interaction (MobileHCI '24). 8, MHCI, Article 277, 27 pages. <https://doi.org/10.1145/3676524>. **Best Paper Award**

[C.4] Momona Yamagami, Alexandra A. Portnova-Fahreeva, **Junhan Kong**, Jacob O. Wobbrock, Jennifer Mankoff. How Do People with Limited Movement Personalize Upper-Body Gestures? Considerations for the Design of Personalized and Accessible Gesture Interfaces. In Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23). Association for Computing Machinery, New York, NY, USA, Article 1, 1–15. <https://doi.org/10.1145/3597638.3608430>.



[C.3] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. Quantifying Touch: New Metrics for Characterizing What Happens *During* a Touch. In The 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22), October 23–26, 2022, Athens, Greece. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3517428.3544804>. **Best Paper Nominee**

[C.2] **Junhan Kong**, Dena Sabha, Jeffrey P. Bigham, Amy Pavel, Anhong Guo. 2021. TutorialLens: Authoring Interactive Augmented Reality Tutorials Through Narration and Demonstration. In Symposium on Spatial User Interaction (SUI '21). Association for Computing Machinery, New York, NY, USA, Article 16, 1–11. <https://doi.org/10.1145/3485279.3485289>.

[C.1] Anhong Guo, **Junhan Kong**, Michael Rivera, Frank F. Xu, Jeffrey P. Bigham. 2019. StateLens: A Reverse Engineering Solution for Making Existing Dynamic Touchscreens Accessible. In Proceedings of the 32nd

Annual ACM Symposium on User Interface Software and Technology (UIST '19). Association for Computing Machinery, New York, NY, USA, 371–385. <https://doi.org/10.1145/3332165.3347873>.

Posters and Demos

[P.4] Momona Yamagami, Claire L. Mitchell, Alexandra A Portnova-Fahreeva, **Junhan Kong**, Jacob O. Wobbrock. (2024). Personalized gesture classification for encouraging non-sedentary behavior during technology use in people with motor disabilities. Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '24 Poster). July 15-19, 2024, Orlando, Florida. Piscataway, New Jersey: IEEE Engineering in Medicine and Biology Society.

[P.3] **Junhan Kong**, Tianyuan Cai, Zoya Bylinskii. Improving Mobile Reading Experiences while Walking Through Automatic Adaptations and Prompted Customization. In Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Poster). Association for Computing Machinery, New York, NY, USA, Article 19, 1–3. <https://doi.org/10.1145/3586182.3616666>.

[P.2] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. 2021. New Metrics for Understanding Touch by People with and without Limited Fine Motor Function. In The 23rd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '21 Poster). Association for Computing Machinery, New York, NY, USA, Article 80, 1–4. <https://doi.org/10.1145/3441852.3476559>.

[P.1] **Junhan Kong**, Anhong Guo, Jeffrey P. Bigham. 2019. Supporting Older Adults in Using Complex User Interfaces with Augmented Reality. In The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19 Demo). Association for Computing Machinery, New York, NY, USA, 661–663. <https://doi.org/10.1145/3308561.3354593>.

PATENTS

[IP.1] Anhong Guo, **Junhan Kong**, Michael Rivera, Frank F. Xu, Jeffrey P. Bigham. StateLens: A Reverse Engineering Solution for Making Existing Dynamic Touchscreens Accessible. U.S. Provisional Patent Application 19/207, filed June 6, 2019.

INDUSTRY EXPERIENCE

Research Scientist Intern, Meta Reality Labs

Jun 2024 - Sep 2024

Mentors: Amy Karlson, Hemant Surale, Michael Proulx

Devised new metrics for quantitatively characterizing 3D pointing. Developed pointing techniques using gaze, hand, and ring devices and investigated how the metrics characterize pointing performance and user experience using the techniques. Led to full ICMI paper [C.9].

Research Scientist Intern, Adobe

Jun 2023 - Sep 2023

Mentors: Tianyuan Cai, Zoya Bylinskii

Investigated the impact of walking on mobile reading experiences, developed a system that provides automatic and customized reading adaptations to walking using smartphone built-in sensors. Winner of the Intern Project Expo (Research Org). Led to UIST poster [P.3] and full CHI paper [C.6].

Software Engineering Intern, Google

May 2018 - Aug 2018

Software Engineering Intern, Jet.com

Jun 2017 - Aug 2017

TEACHING

Instructor

UW INFO 498 Special Topics in Informatics: Accessibility (Course Rating: 4.8/5.0) Spring 2024

Teaching Assistant

UW INFO 463 Input and Interaction Spring 2025

UW HCID 520 User Interface Software and Technology Winter 2023, 2025, 2026

UW INFO 380 Information Systems Analysis and Design Fall 2020, 2022, 2023, 2025

UW IMT 575 Data Science: Scaling, Applications and Ethics Spring 2022

UW IMT 596 & 597 MSIM Capstone Winter 2021, Spring 2021

CMU 05-391 Designing Human-Centered Software Fall 2017 - Fall 2019

CMU 15-122 Principles of Imperative Computation Fall 2016 - Fall 2019

SERVICE

Organizing Committee

ASSETS 2025 Student Volunteer Co-Chair

ASSETS 2022 Web and Graphics Design Co-Chair

UW DUB Doctoral Colloquium 2023 Coordinator

Conference Reviewer

Special Recognitions: UIST ('23, '24), CHI ('22)

Conference Reviewer:

UIST ('23, '24), CHI ('22, '24, '25, '26), ASSETS ('23 Posters and Demos), SUI ('24), DIS ('24)

Undergraduate Activities

CMU Undergraduate HCI Student Advisory Committee Sep 2018 - May 2019

CMU Undergraduate Orientation Counselor Aug 2018